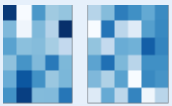
1. Data and QC

GSE193103

raw Salmon counts



125 libraries:

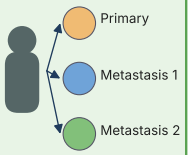
- 100 tumour
- 22 normal
- 3 unclassified

QC removes 1 library (99 tumour kept)

27,809 genes retained

2. Patient units

Libraries averaged per patient × site



15 paired patients (8 preservation-matched)

12 patients with ≥ 2 sites

x = FFPE share difference (metastasis − primary) used by the preservation-aware model

3. Analysis axes

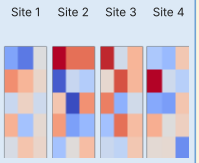
Axis A: state

within patient, paired: metastasis − primary



Axis B: site

within patient, blocked F across metastatic sites



site variation, patient structure kept

4. Layers S1–S3, S5

S1. Patient-level stability selection

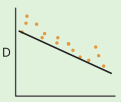
- complementary patient half-samples
- selection frequency π ≥ 0.8; top q = 100



S2. Preservation-aware estimation

$D = \beta + \delta x + \varepsilon$

β: state effect at x = 0



S3. Replicate reproducibility

ICC ≥ 0.5 (replicates: 14 of 82 units)

S5. Host-organ correction

- normal liver, lung, brain signatures
- host fraction per unit; rerun
- organ-enriched: excess ≥ 2 $\log_2$
- same sign in ≥ 3 of 4 sites

5. Calibration (S4)

(patient-structured nulls)

rerun the whole pipeline (S1–S3, S5) on null data

Sign-flip null (Axis A, agnostic)



Freedman–Lane null (aware model)



Within-patient site permutation null (Axis B)



MB bound also reported; nulls assess calibration and are not relied on alone

6. Output

Robust feature =

passes all layers and beats its null

This cohort, among the 367 conventional hits (not nested):

- **367** conventional significant
- **11** stable under patient resampling
- **1** preservation-aware stable
- **0** robust

S4 reruns the pipeline (S1–S3, S5) on null data to calibrate its selection counts